

COVID Confounding

In August 2021, the UK, in Technical Bulletin #20, put out their latest statistics on the Delta variant, reporting the number of positive cases and deaths (classified by vaccination status and age).

Lumping the two age groups together (under 50 and above 50), it initially seems as if those that are vaccinated are at twice the risk of dying if they contract Delta:

	A	B	C	D
1	Population		All reported positive	cases that are ID as Delta
2	Group		Some level of vaccination	Unvaccinated
3	Cases		117115	151054
4	Deaths		481	253
5	Mortality rate		0.41%	0.17%
6	Risk Ratio (Vacc/Unvacc)		245.2%	

However, breaking down the cases by age group (the confounding variable here), we come to the opposite (and correct) conclusion: those that are vaccinated are much less likely to die if they contract Delta:

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8	Population		All reported positive	cases that are ID as Delta
9	Group		Some level of vaccination (<50)	Unvaccinated
10	Cases		89807	147612
11	Deaths		21	48
12	Mortality rate		0.02%	0.03%
13	Risk Ratio (Vacc/Unvacc)		71.9%	
14				
15	Population		All reported positive	cases that are ID as Delta
16	Group		Some level of vaccination (>=50)	Unvaccinated
17	Cases		27307	3440
18	Deaths		460	205
19	Mortality rate		1.68%	5.96%
20	Risk Ratio (Vacc/Unvacc)		28.3%	